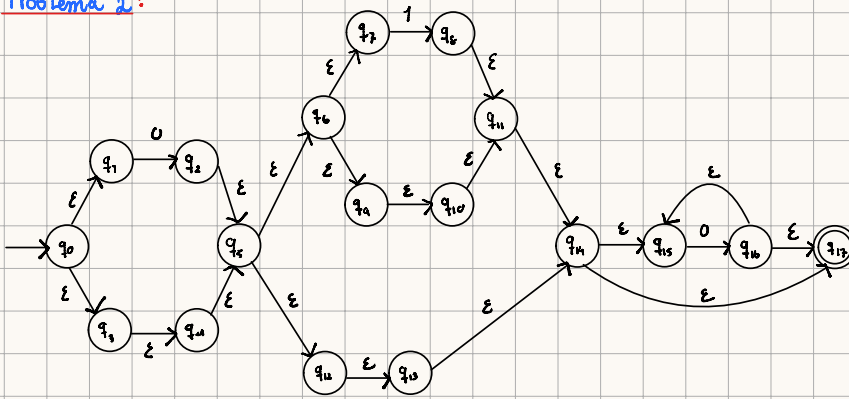


Problema 2:



$$q_0 = \epsilon$$

$$q_1 = q_0 \epsilon$$

$$q_2 = q_1 0 = q_0 \epsilon 0 = \epsilon \epsilon 0 = 0$$

$$q_3 = q_0 \epsilon$$

$$q_4 = q_3 \epsilon = q_0 \epsilon \epsilon = \epsilon \epsilon = \epsilon$$

$$q_5 = q_2 \epsilon + q_4 \epsilon = 0 + \epsilon$$

$$q_6 = q_5 \epsilon = (0 + \epsilon) \epsilon = 0 + \epsilon$$

$$q_7 = q_6 \epsilon = (0 + \epsilon) \epsilon \epsilon = 0 + \epsilon$$

$$q_8 = q_7 1 = (0 + \epsilon) \epsilon \epsilon \cdot 1 = (0 + \epsilon) 1$$

$$q_9 = q_8 \epsilon = (0 + \epsilon) \epsilon \epsilon \epsilon = (0 + \epsilon)$$

$$q_{10} = q_9 \epsilon = (0 + \epsilon) \epsilon \epsilon \epsilon \epsilon = (0 + \epsilon)$$

$$q_{11} = q_8 \epsilon + q_{10} \epsilon = (0 + \epsilon) \epsilon \epsilon \cdot 1 + (0 + \epsilon) \epsilon \epsilon \epsilon \epsilon = (0 + \epsilon) 1 + (0 + \epsilon) = (0 + \epsilon)(1 + \epsilon)$$

$$q_{12} = q_5 \epsilon = (0 + \epsilon) \epsilon = (0 + \epsilon)$$

$$q_{13} = q_{12} \epsilon = (0 + \epsilon) \epsilon = (0 + \epsilon)$$

$$q_{14} = q_{11} \epsilon + q_{13} \epsilon = (0 + \epsilon) \epsilon \epsilon \cdot 1 + (0 + \epsilon) \epsilon \epsilon \epsilon \epsilon + (0 + \epsilon) \epsilon = (0 + \epsilon)(1 + \epsilon) + (0 + \epsilon) = (0 + \epsilon)(1 + \epsilon + \epsilon) = (0 + \epsilon)(1 + \epsilon)$$

$$q_{15} = q_{14} \epsilon + q_{10} \epsilon$$

$$q_{16} = q_{15} 0 = (q_{14} \epsilon + q_{10} \epsilon) 0 = q_{14} \epsilon 0 + q_{10} \epsilon 0 = (0 + \epsilon)(1 + \epsilon) 0 + q_{10} 0 = (0 + \epsilon)(1 + \epsilon) 0^* 0$$

$$q_{17} = q_{16} \epsilon + q_{14}$$

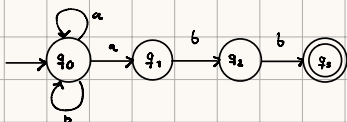
$$q_{17} = (0 + \epsilon)(1 + \epsilon) 0^* + (0 + \epsilon)(1 + \epsilon)$$

$$q_{17} = (0 + \epsilon)(1 + \epsilon)(\epsilon + 0^* 0)$$

$$q_{17} = (0 + \epsilon)(1 + \epsilon) 0^*$$

$$R = (0 + \epsilon)(1 + \epsilon) 0^*$$

Ejercicios Extra:



$$q_0 = \epsilon + q_0 a + q_0 b$$

$$q_0 = \epsilon + q_0 (a + b)$$

$$q_0 = \epsilon (a + b)^*$$

$$q_0 = (a + b)^*$$

$$q_2 = q_1 b$$

$$q_2 = (a + b)^* a b$$

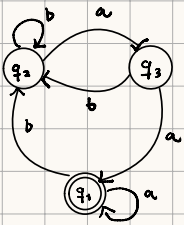
$$q_3 = q_2 b$$

$$q_3 = (a + b)^* a b b$$

$$q_1 = q_0 a$$

$$q_1 = (a + b)^* a$$

$$R = (a + b)^* a b b$$



$$q_3 = q_2 a$$

$$q_2 = q_2 b + q_3 b + q_1 b$$

$$q_1 = q_1 a + q_3 a$$

$$q_1 = q_1 a + q_1 a b (a b + b)^*$$

$$q_1 = q_1 (a + [a a b (a b + b)^*]^*) + \epsilon$$

$$q_1 = (a + [a a b (a b + b)^*]^*)^*$$

$$q_2 = q_2 b + q_2 a b + q_1 b$$

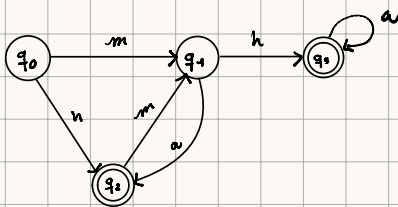
$$q_2 = q_2 (b + a b) + q_1 b$$

$$q_2 = q_1 b (b + a b)^*$$

$$q_3 = q_1 a b (a b + b)^*$$

$$q_3 = a a b (a b + b)^* q_1$$

$$R = (a + [a a b (a b + b)^*]^*)^*$$



$$q_0 = \epsilon$$

$$q_1 = q_0 m + q_2 m$$

$$q_1 = m + q_2 m$$

$$q_1 = m + (h + q_1 a) m$$

$$q_1 = m + h m + q_1 a m$$

$$q_1 = (m + h m) (a m)^*$$

$$q_2 = q_0 h + q_1 a$$

$$q_2 = h + (m + h m) (a m)^* a$$

$$R = (h + (m + h m) (a m)^* a) ((m + h m) (a m)^* a)^*$$

$$q_3 = q_1 h + q_3 a$$

$$q_3 = (m + h m) (a m)^* + q_3 a$$

$$q_3 = (m + h m) (a m)^* a^*$$